

```
!pip install -r ./requirements.txt
```



```
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from -r ./requirements.txt (line 1)) (2.0.2)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from -r ./requirements.txt (line 2)) (2.2.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from -r ./requirements.txt (line 3)) (3.10.0)
Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (from -r ./requirements.txt (line 4)) (4.12.0.88)
Requirement already satisfied: Pillow in /usr/local/lib/python3.12/dist-packages (from -r ./requirements.txt (line 5)) (11.3.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (from -r ./requirements.txt (line 6)) (2.8.0+cu126)
Requirement already satisfied: diffusers in /usr/local/lib/python3.12/dist-packages (from -r ./requirements.txt (line 7)) (0.35.2)
Collecting autodistill (from -r ./requirements.txt (line 8))
  Downloading autodistill-0.1.29-py3-none-any.whl.metadata (32 kB)
Collecting autodistill-grounding-dino (from -r ./requirements.txt (line 9))
  Downloading autodistill_grounding_dino-0.1.4-py3-none-any.whl.metadata (3.0 kB)
Requirement already satisfied: openai in /usr/local/lib/python3.12/dist-packages (from -r ./requirements.txt (line 10)) (1.109.1)
Collecting autodistill-yolov8 (from -r ./requirements.txt (line 11))
  Downloading autodistill_yolov8-0.1.4-py3-none-any.whl.metadata (995 bytes)
Collecting roboflow (from -r ./requirements.txt (line 12))
  Downloading roboflow-1.2.11-py3-none-any.whl.metadata (9.7 kB)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->-r ./requirements.txt (line 2)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->-r ./requirements.txt (line 2)) (2025.2
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Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 3))
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Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 3))
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Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 3))
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 6)) (3.20.0)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 6))
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 6)) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 6)) (1.13.3
Requirement already satisfied: networkx in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 6)) (3.5)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 6)) (3.1.6)
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Requirement already satisfied: NVIDIA-CUDA-12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt
Requirement already satisfied: NVIDIA-CUDA-12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt
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Requirement already satisfied: NVIDIA-CUDNN-9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (l
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Requirement already satisfied: NVIDIA-CURAND-10.3.7.77 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (
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Requirement already satisfied: NVIDIA-CUSPARSE-12.5.4.2 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt
Requirement already satisfied: NVIDIA-CUSPARSELT-0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (
Requirement already satisfied: NVIDIA-NCCL-2.27.3 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
Requirement already satisfied: NVIDIA-NVTX-12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
Requirement already satisfied: NVIDIA-NVJITLINK-12.6.85 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt
Requirement already satisfied: NVIDIA-CUFILE-1.11.1.6 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (l
Requirement already satisfied: triton==3.4.0 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 6)) (3.4.0)
Requirement already satisfied: importlib_metadata in /usr/local/lib/python3.12/dist-packages (from diffusers->-r ./requirements.txt (line 7)
Requirement already satisfied: huggingface-hub>=0.34.0 in /usr/local/lib/python3.12/dist-packages (from diffusers->-r ./requirements.txt (li
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from diffusers->-r ./requirements.txt (line 7))
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Requirement already satisfied: safetensors>=0.3.1 in /usr/local/lib/python3.12/dist-packages (from diffusers->-r ./requirements.txt (line 7)
Collecting supervision (from autodistill->-r ./requirements.txt (line 8))
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Collecting supervision (from autodistill->-r ./requirements.txt (line 8))
  Downloading supervision-0.26.1-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from autodistill->-r ./requirements.txt (line 8)) (4.67.1)
Requirement already satisfied: PyYAML==5.3.1 in /usr/local/lib/python3.12/dist-packages (from autodistill->-r ./requirements.txt (line 8)) (
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from autodistill->-r ./requirements.txt (line 8)) (8.3.0)
Collecting rf-groundingdino==0.1.2 (from autodistill-grounding-dino->-r ./requirements.txt (line 9))
  Downloading rf_groundingdino-0.1.2.tar.gz (82 kB)
      82.4/82.4 kB 6.2 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Collecting rf-segment-anything (from autodistill-grounding-dino->-r ./requirements.txt (line 9))
  Downloading rf_segment_anything-1.0-py3-none-any.whl.metadata (583 bytes)
Requirement already satisfied: torchvision in /usr/local/lib/python3.12/dist-packages (from rf-groundingdino==0.1.2->autodistill-grounding-d
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (from rf-groundingdino==0.1.2->autodistill-grounding-
Collecting addict (from rf-groundingdino==0.1.2->autodistill-grounding-dino->-r ./requirements.txt (line 9))
  Downloading addict-2.4.0-py3-none-any.whl.metadata (1.0 kB)
Collecting yapf (from rf-groundingdino==0.1.2->autodistill-grounding-dino->-r ./requirements.txt (line 9))
  Downloading yapf-0.43.0-py3-none-any.whl.metadata (46 kB)
      46.8/46.8 kB 4.7 MB/s eta 0:00:00
Requirement already satisfied: timm in /usr/local/lib/python3.12/dist-packages (from rf-groundingdino==0.1.2->autodistill-grounding-dino->-r
Requirement already satisfied: pycocotools in /usr/local/lib/python3.12/dist-packages (from rf-groundingdino==0.1.2->autodistill-grounding-d
Requirement already satisfied: anyio<5,>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from openai->-r ./requirements.txt (line 10)) (4.
Requirement already satisfied: distro<2,>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from openai->-r ./requirements.txt (line 10)) (1.
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from openai->-r ./requirements.txt (line 10)) (0.
Requirement already satisfied: jiter<1,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from openai->-r ./requirements.txt (line 10)) (0.
Requirement already satisfied: pydantic<3,>=1.9.0 in /usr/local/lib/python3.12/dist-packages (from openai->-r ./requirements.txt (line 10))
Requirement already satisfied: sniffio in /usr/local/lib/python3.12/dist-packages (from openai->-r ./requirements.txt (line 10)) (1.3.1)
Collecting ultralytics==8.0.81 (from autodistill-yolov8->-r ./requirements.txt (line 11))
  Downloading ultralytics-8.0.81-py3-none-any.whl.metadata (25 kB)
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics==8.0.81->autodistill-yolov8->-r ./r
Requirement already satisfied: seaborn>=0.11.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics==8.0.81->autodistill-yolov8->-r
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from ultralytics==8.0.81->autodistill-yolov8->-r ./require
Collecting thop>=0.1.1 (from ultralytics==8.0.81->autodistill-yolov8->-r ./requirements.txt (line 11))
  Downloading thop-0.1.1.post2209072238-py3-none-any.whl.metadata (2.7 kB)
Requirement already satisfied: sentry-sdk in /usr/local/lib/python3.12/dist-packages (from ultralytics==8.0.81->autodistill-yolov8->-r ./req
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from roboflow->-r ./requirements.txt (line 12)) (2025.10.
Collecting idna==3.7 (from roboflow->-r ./requirements.txt (line 12))
  Downloading idna-3.7-py3-none-any.whl.metadata (9.9 kB)
Collecting opencv-python-headless==4.10.0.84 (from roboflow->-r ./requirements.txt (line 12))
  Downloading opencv_python_headless-4.10.0.84-cp37-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (20 kB)
Collecting pi-heif<2 (from roboflow->-r ./requirements.txt (line 12))
  Downloading pi_heif-1.1.1-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (6.5 kB)
Collecting pillow-avif-plugin<2 (from roboflow->-r ./requirements.txt (line 12))
  Downloading pillow_avif_plugin-1.5.2-cp312-cp312-manylinux_2_28_x86_64.whl.metadata (2.1 kB)
Requirement already satisfied: python-dotenv in /usr/local/lib/python3.12/dist-packages (from roboflow->-r ./requirements.txt (line 12)) (1.
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages (from roboflow->-r ./requirements.txt (line 12)) (1.17.0)
Requirement already satisfied: urllib3>=1.26.6 in /usr/local/lib/python3.12/dist-packages (from roboflow->-r ./requirements.txt (line 12)) (
Requirement already satisfied: requests-toolbelt in /usr/local/lib/python3.12/dist-packages (from roboflow->-r ./requirements.txt (line 12))
Collecting filetype (from roboflow->-r ./requirements.txt (line 12))
  Downloading filetype-1.2.0-py2.py3-none-any.whl.metadata (6.5 kB)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->openai->-r ./requirements.tx
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->openai->-r ./requ
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.34.0->diffusers->-r
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1.9.0->openai->-r ./requ
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Requirement already satisfied: pydantic-core==2.33.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1.9.0->openai->-r ./requirements.txt)
Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1.9.0->openai->-r ./requirements.txt)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->diffusers->-r ./requirements.txt)
Requirement already satisfied: defusedxml>=0.7.1 in /usr/local/lib/python3.12/dist-packages (from supervision->autodistill->-r ./requirements.txt)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch->-r ./requirements.txt)
Requirement already satisfied: zipp>=3.20 in /usr/local/lib/python3.12/dist-packages (from importlib_metadata->diffusers->-r ./requirements.txt)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch->-r ./requirements.txt (line 6))
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers->rf-groundingdino==0.1.2->autodistill)
Requirement already satisfied: platformdirs>=3.5.1 in /usr/local/lib/python3.12/dist-packages (from yapf->rf-groundingdino==0.1.2->autodistill)
Downloading autodistill-0.1.29-py3-none-any.whl (32 kB)
Downloading autodistill_grounding_dino-0.1.4-py3-none-any.whl (9.1 kB)
Downloading autodistill_yolov8-0.1.4-py3-none-any.whl (14 kB)
Downloading ultralytics-8.0.81-py3-none-any.whl (527 kB)
 257.0/527.0 kB 21.0 MB/s eta 0:00:00
Downloading roboflow-1.2.11-py3-none-any.whl (89 kB)
 89.9/89.9 kB 10.6 MB/s eta 0:00:00
Downloading idna-3.7-py3-none-any.whl (66 kB)
 66.8/66.8 kB 7.1 MB/s eta 0:00:00
Downloading opencv_python_headless-4.10.0.84-cp37-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (49.9 MB)
 49.9/49.9 MB 16.8 MB/s eta 0:00:00
Downloading pi_heif-1.1.1-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (1.4 MB)
 1.4/1.4 MB 72.9 MB/s eta 0:00:00
Downloading pillow_avif_plugin-1.5.2-cp312-cp312-manylinux_2_28_x86_64.whl (4.2 MB)
 4.2/4.2 MB 108.5 MB/s eta 0:00:00
Downloading supervision-0.26.1-py3-none-any.whl (207 kB)
 207.2/207.2 kB 17.6 MB/s eta 0:00:00
Downloading filetype-1.2.0-py2.py3-none-any.whl (19 kB)
Downloading rf_segment_anything-1.0-py3-none-any.whl (36 kB)
Downloading thop-0.1.1.post2209072238-py3-none-any.whl (15 kB)
Downloading addict-2.4.0-py3-none-any.whl (3.8 kB)
Downloading yapf-0.43.0-py3-none-any.whl (256 kB)
 256.2/256.2 kB 22.2 MB/s eta 0:00:00
Building wheels for collected packages: rf-groundingdino
  Building wheel for rf-groundingdino (setup.py) ... done
  Created wheel for rf-groundingdino: filename=rf_groundingdino-0.1.2-py2.py3-none-any.whl size=88734 sha256=1350fd6c459330b1924815600ee27ca
  Stored in directory: /root/.cache/pip/wheels/e1/4a/ca/86ae3ab93b9e04d9aecd76140b62cd0931d0aea6710a266ae9
Successfully built rf-groundingdino
Installing collected packages: rf-segment-anything, pillow-avif-plugin, filetype, addict, yapf, pi-heif, opencv-python-headless, idna, supervision
  Attempting uninstall: opencv-python-headless
    Found existing installation: opencv-python-headless 4.12.0.88
    Uninstalling opencv-python-headless-4.12.0.88:
      Successfully uninstalled opencv-python-headless-4.12.0.88
  Attempting uninstall: idna
    Found existing installation: idna 3.11
    Uninstalling idna-3.11:
      Successfully uninstalled idna-3.11
Successfully installed addict-2.4.0 autodistill-0.1.29 autodistill-grounding-dino-0.1.4 autodistill-yolov8-0.1.4 filetype-1.2.0 idna-3.7 opencv-python-headless-4.10.0.84 pi-heif-1.1.1 pillow-avif-plugin-1.5.2 rf-groundingdino-0.1.2 rf-segment-anything-1.0 supervision-0.26.1 thop-0.1.1.post2209072238 yapf-0.43.0
WARNING: The following packages were previously imported in this runtime:
[cv2,idna]
You must restart the runtime in order to use newly installed versions.

```

RESTART SESSION

```
!pip install google-generativeai
```

```
Requirement already satisfied: google-generativeai in /usr/local/lib/python3.12/dist-packages (0.8.5)
Requirement already satisfied: google-ai-generativelanguage==0.6.15 in /usr/local/lib/python3.12/dist-packages (from google-generativeai) (0.6.15)
Requirement already satisfied: google-api-core in /usr/local/lib/python3.12/dist-packages (from google-generativeai) (2.27.0)
Requirement already satisfied: google-api-python-client in /usr/local/lib/python3.12/dist-packages (from google-generativeai) (2.185.0)
Requirement already satisfied: google-auth>=2.15.0 in /usr/local/lib/python3.12/dist-packages (from google-generativeai) (2.38.0)
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from google-generativeai) (5.29.5)
Requirement already satisfied: pydantic in /usr/local/lib/python3.12/dist-packages (from google-generativeai) (2.11.10)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from google-generativeai) (4.67.1)
Requirement already satisfied: typing-extensions in /usr/local/lib/python3.12/dist-packages (from google-generativeai) (4.15.0)
Requirement already satisfied: proto-plus<2.0.0dev,>=1.22.3 in /usr/local/lib/python3.12/dist-packages (from google-ai-generativelanguage==0.6.15->google-generativeai) (1.24.0)
Requirement already satisfied: googleapis-common-protos<2.0.0,>=1.56.2 in /usr/local/lib/python3.12/dist-packages (from google-api-core->google-generativeai) (1.66.0)
Requirement already satisfied: requests<3.0.0,>=2.18.0 in /usr/local/lib/python3.12/dist-packages (from google-api-core->google-generativeai) (2.32.0)
Requirement already satisfied: cachetools<6.0,>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from google-auth>=2.15.0->google-generativeai) (5.5.2)
Requirement already satisfied: pyasn1-modules>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from google-auth>=2.15.0->google-generativeai) (0.6.1)
Requirement already satisfied: rsa<5,>=3.1.4 in /usr/local/lib/python3.12/dist-packages (from google-auth>=2.15.0->google-generativeai) (4.9.1)
Requirement already satisfied: httplib2<1.0.0,>=0.19.0 in /usr/local/lib/python3.12/dist-packages (from google-api-python-client->google-generativeai) (0.11.0)
Requirement already satisfied: google-auth-httplib2<1.0.0,>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from google-api-python-client->google-generativeai) (0.1.0)
Requirement already satisfied: uritemplate<5,>=3.0.1 in /usr/local/lib/python3.12/dist-packages (from google-api-python-client->google-generativeai) (4.1.2)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic->google-generativeai) (0.7.0)
Requirement already satisfied: pydantic-core==2.33.2 in /usr/local/lib/python3.12/dist-packages (from pydantic->google-generativeai) (2.33.2)
Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from pydantic->google-generativeai) (0.4.2)
Requirement already satisfied: grpcio<2.0.0,>=1.33.2 in /usr/local/lib/python3.12/dist-packages (from google-api-core[grpc]!=2.0.*,!=2.1.*,!=2.2.*->google-generativeai) (1.66.2)
Requirement already satisfied: grpcio-status<2.0.0,>=1.33.2 in /usr/local/lib/python3.12/dist-packages (from google-api-core[grpc]!=2.0.*,!=2.1.*,!=2.2.*->google-generativeai) (1.66.2)
Requirement already satisfied: pyparsing<4,>=3.0.4 in /usr/local/lib/python3.12/dist-packages (from httplib2<1.0.0,>=0.19.0->google-api-python-client->google-generativeai) (3.1.4)
Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in /usr/local/lib/python3.12/dist-packages (from pyasn1-modules>=0.2.1->google-auth>=2.15.0->google-generativeai) (0.6.1)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.18.0->google-api-core->google-generativeai) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.18.0->google-api-core->google-generativeai) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.18.0->google-api-core->google-generativeai) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.18.0->google-api-core->google-generativeai) (2025.1.31)
```

```
import os
import math
import io
import ast
import base64
from io import BytesIO

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

import cv2
from PIL import ImageDraw

import torch
from diffusers import StableDiffusionPipeline
from autodistill.detection import CaptionOntology
```



```
from autodistill_grounding_dino import GroundingDINO
import google.generativeai as genai
```

Importing from `timm.models.layers` is deprecated, please import via `timm.layers`

```
important_objects = "an antique brass astrolabe"
number_of_prompts = 50

description_of_prompt = "A detailed photo of astrolabe on various surfaces like a wooden desk, a velvet cloth, or held in hands, with different backgrounds."

base_prompt = f'''
# List of Important Objects:
# The objects listed here must be included in every generated prompt.
Important Objects that must be present in each prompt:
{important_objects}

# Input Details:
# The task is to generate a specific number of prompts related to the description provided.
Input:
Generate {number_of_prompts} realistic prompts related to {description_of_prompt} for image generation.

# Instructions for Prompt Generation:
# - Each prompt should depict real-life behaviors and scenarios involving the objects.
# - All important objects should be included in every prompt.
# - Ensure that the objects are captured at varying distances from the camera:
#   - From very close-up shots to objects in the far background.
# - The prompts should be diverse and detailed to cover a wide range of use cases.

# Output Format:
# - The output should be a Python list containing all the generated prompts as strings.
# - Each prompt should be enclosed in quotation marks and separated by commas within the list.
Output:
Return a Python list containing these prompts as strings for later use in training a computer vision model.
[prompt1, prompt2, ...]
'''

print(base_prompt)
```

```
# List of Important Objects:
# The objects listed here must be included in every generated prompt.
Important Objects that must be present in each prompt:
an antique brass astrolabe

# Input Details:
# The task is to generate a specific number of prompts related to the description provided.
Input:
Generate 50 realistic prompts related to A detailed photo of astrolabe on various surfaces like a wooden desk, a velvet cloth, or held in hands, with different backgrounds.
```

```
# Instructions for Prompt Generation:
# - Each prompt should depict real-life behaviors and scenarios involving the objects.
# - All important objects should be included in every prompt.
# - Ensure that the objects are captured at varying distances from the camera:
#   - From very close-up shots to objects in the far background.
# - The prompts should be diverse and detailed to cover a wide range of use cases.

# Output Format:
# - The output should be a Python list containing all the generated prompts as strings.
# - Each prompt should be enclosed in quotation marks and separated by commas within the list.
Output:
Return a Python list containing these prompts as strings for later use in training a computer vision model.
[prompt1, prompt2, ...]
```

```
try:
    genai.configure(api_key="AIzaSyAF6HZ27Wl5qyMigUy85zXPNmOK-ckHuls")
except Exception as e:
    print(f"Error: Could not configure Gemini. Please ensure your API key is set correctly. Details: {e}")

model = genai.GenerativeModel('gemma-3-1b-it')

full_prompt = f"""{base_prompt}"

your response: [prompt1, prompt2, ...] and do not say anything else and i will be using ast.literal_eval to convert the string to a list"

try:
    completion = model.generate_content(full_prompt)
    response_text = completion.text

except Exception as e:
    print(f"An error occurred while calling the Gemini API: {e}")
    response_text = "[]"

prompts_list = ast.literal_eval(response_text)
print(prompts_list)
```


Traceback (most recent call last):

```
File "/usr/local/lib/python3.12/dist-packages/IPython/core/interactiveshell.py", line 3553, in run_code
    exec(code_obj, self.user_global_ns, self.user_ns)

File "/tmp/ipython-input-330894657.py", line 21, in <cell line: 0>
    prompts_list = ast.literal_eval(response_text)

File "/usr/lib/python3.12/ast.py", line 66, in literal_eval
    node_or_string = parse(node_or_string.lstrip(" \t"), mode='eval')

File "/usr/lib/python3.12/ast.py", line 52, in parse
    return compile(source, filename, mode, flags,
                  ^
SyntaxError: invalid character "'" (U+201C)
```

Next steps: [Explain error](#)

```
import re

variable_definition = response_text.strip()

# Use regex to find the list-like string
match = re.search(r'\[.*?\]', variable_definition, re.DOTALL)
if match:
    list_string = match.group(0)
else:
    list_string = "[]" # Default to empty list if no list is found

prompts = ast.literal_eval(list_string)

print(prompts[0:5])
```

```
[]
```

```
# prompts = prompts * 2

import random
random.shuffle(prompts)

len(prompts)
```

```
import torch

if torch.cuda.is_available():
    print("Success! PyTorch can see your GPU.")
    print(f"Device name: {torch.cuda.get_device_name(0)}")
else:
    print("Failure. PyTorch cannot see your GPU. Check your NVIDIA drivers and installation.")
```

Success! PyTorch can see your GPU.
Device name: Tesla T4

```
model_id = "CompVis/stable-diffusion-v1-4"
device = "cuda"
# pkl_filename = "stable_diffusion_pipe.pkl"

pipe = StableDiffusionPipeline.from_pretrained(model_id, torch_dtype=torch.float16)
pipe = pipe.to(device)
```

The secret `HF\_TOKEN` does not exist in your Colab secrets.

To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it as secret in y
You will be able to reuse this secret in all of your notebooks.

Please note that authentication is recommended but still optional to access public models or datasets.

Loading pipeline components...: 100% 7/7 [00:27<00:00, 3.03s/it]

`torch\_dtype` is deprecated! Use `dtype` instead!

```
import pickle
```

```
print(f"\nAttempting to save pipeline to {pkl_filename} using pickle...")
pipe.to("cpu")
```

```
with open(pkl_filename, 'wb') as f:
    pickle.dump(pipe, f)
```

```
print("Pickle save complete.")
```

```
# --- How to load it back ---
print("\nLoading pipeline from pickle file...")
with open(pkl_filename, 'rb') as f:
    reloaded_pipe = pickle.load(f)
```

```
# Move the reloaded pipe back to the GPU to use it
reloaded_pipe = reloaded_pipe.to(device)
print("Pipeline reloaded from pickle and moved to CUDA.")
```

Attempting to save pipeline to stable_diffusion_pipe.pkl using pickle...

Pipelines loaded with `dtype=torch.float16` cannot run with `cpu` device. It is not recommended to move them to `cpu` as running them will fail.

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Pipelines loaded with `dtype=torch.float16` cannot run with `cpu` device. It is not recommended to move them to `cpu` as running them will fail.

Pickle save complete.

Loading pipeline from pickle file...

Pipeline reloaded from pickle and moved to CUDA.

```
sample_prompts = prompts[:10]
images = pipe(sample_prompts).images
```

```
synthetic_data = pd.DataFrame({'Prompt': sample_prompts, 'Image': images})
synthetic_data
```

Let's visualize the generated images and see how they look:

```
def display_images(dataframe):
    images = dataframe['Image']
    max_images_per_row = 5
    rows = (len(images) + max_images_per_row - 1) // max_images_per_row
    columns = min(len(images), max_images_per_row)

    fig, axes = plt.subplots(rows, columns, figsize=(columns * 5, rows * 5))

    if rows == 1:
        axes = [axes]
    else:
        axes = axes.flatten()
    for i, image in enumerate(images):
        axes[i].imshow(image)
        axes[i].axis('off')

    for j in range(i + 1, len(axes)):
        axes[j].axis('off')

    plt.show()

display_images(synthetic_data)
```



```
import supervision as sv
from autodistill.detection import CaptionOntology
from autodistill_grounding_dino import GroundingDINO
from PIL import Image
import numpy as np
import cv2

ontology = CaptionOntology({
    "car": "black car"
})

grounding_dino = GroundingDINO(ontology=ontology)

annotated_images = []
detections_list = []

box_annotator = sv.BoxAnnotator(thickness=2)
label_annotator = sv.LabelAnnotator(text_thickness=1, text_scale=0.5)

print("Starting auto-annotation process...")

for index, row in synthetic_data.iterrows():
    pil_image = row['Image']
```

```
pil_image = row[ image ]

# Convert PIL Image to an OpenCV-compatible format (numpy array)
image_bgr = cv2.cvtColor(np.array(pil_image), cv2.COLOR_RGB2BGR)

# Run prediction
detections = grounding_dino.predict(image_bgr)
detections_list.append(detections)

# Annotate the image with boxes and labels for visualization
annotated_image = box_annotator.annotate(scene=image_bgr.copy(), detections=detections)
annotated_image = label_annotator.annotate(scene=annotated_image, detections=detections)

# Convert back to PIL Image format for easy display
annotated_pil_image = Image.fromarray(cv2.cvtColor(annotated_image, cv2.COLOR_BGR2RGB))
annotated_images.append(annotated_pil_image)

print(f"Processed image {index+1}/{len(synthetic_data)}. Found {len(detections)} cars.")
synthetic_data['Detections'] = detections_list
synthetic_data['Annotated_Image'] = annotated_images

print("\nAuto-annotation complete!")
```

```
trying to load grounding dino directly
downloading dino model weights
torch.meshgrid: in an upcoming release, it will be required to pass the indexing argument. (Triggered internally at /pytorch/aten/src/ATen/native/TensorIndexing.cpp:100%)
final text_encoder_type: bert-base-uncased
```

```
# Create a new display function for the annotated images
def display_annotated_images(dataframe):
    # Reset the index of the dataframe to ensure sequential indexing
    dataframe = dataframe.reset_index(drop=True)

    images = dataframe['Annotated_Image']
    prompts = dataframe['Prompt']
    max_images_per_row = 3 # Use fewer columns to see details better
    rows = (len(images) + max_images_per_row - 1) // max_images_per_row
    columns = min(len(images), max_images_per_row)

    fig, axes = plt.subplots(rows, columns, figsize=(columns * 7, rows * 7))
    axes = axes.flatten()

    for i, image in enumerate(images):
        axes[i].imshow(image)
        axes[i].set_title(f"Prompt: {prompts[i][:50]}...", fontsize=8) # Show start of prompt
        axes[i].axis('off')

    for j in range(i + 1, len(axes)):
        axes[j].axis('off')

    plt.tight_layout()
    plt.show()

display_annotated_images(synthetic_data)
```